

Question 1: what did you do?

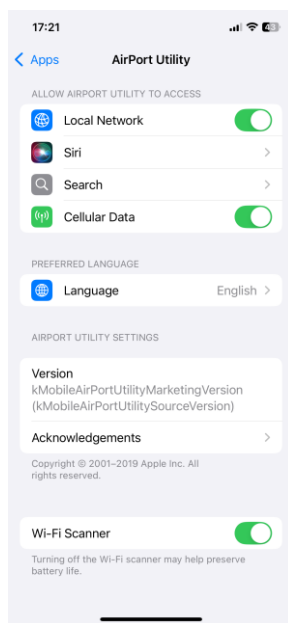
Provide a description of what you did. In particular, I want to know what you did to research the subject of the report. In most cases the value of the research is in the journey more than the result, same here, the better/clearer/insightful your write up is the better it scores.

To start this assignment, I read through the instructions, watched the chapter 3 video and read through the support slide. I then started the work on the assignment by going to the phishing tests, starting with the SonicWall phishing test and then taking the OpenDNS phishing quiz. Once I completed each, I took screenshots of the results and added them to the paper. I then went back to the site to analyse the results and determine where I went wrong and why. While I performed this analysis, there were a few questions that came up, which I researched while performing the analysis. My research process starts by going to Google Gemini and asking this Large Language Model my questions. I then explored the sites that Gemini used to gather its information to provide an answer to determine their reliability, after which I did my own research to expand upon the basic that Gemini provided. Once I completed the phishing portion of this, I then started the Wireshark portion.

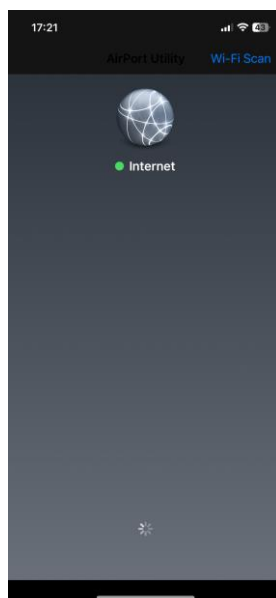
I started the Wireshark section by downloading Wireshark from the link that was provided. I downloaded the version that was designed for a Mac environment, then opened Wireshark. When I opened it, the page had several notifications that came up stating that I needed to also download the CHmodBPF package. I clicked the link to download this package and did so. I was then able to run Wireshark. I ran Wireshark for 30 seconds and saved the captured data under "Wireshark_2807SimkinsLn_Capture." I also opened and reviewed a similar project I had done in CIDM6340 Network Management and Information Security, where Wireshark was taught through Coursera how to use Wireshark and how to analyse the information Wireshark provides. I used this to refresh my memory on Wireshark, as the Coursera was run through a virtual machine, and minimal analysis on my own network was performed.

My next step was to begin my analysis of the scan. There were three things from the prompt that were requested be done. First, a survey of an area (such as a neighbourhood or an apartment complex) was to be conducted to look for unsecured wireless networks. This could be done by looking at the networks detected on a device through the Wi-Fi setting or by using Wireshark. Next, it was requested that some of the open networks that were detected be analysed, or to analyse just your own if you wanted to be safe. It was specifically requested that the protocols and other specific information that Wireshark offers be analysed. The last step was to check how far your own network extends to see if it can be found and accessed

outside of your home. To perform the first request of analysing an area, I used two methods. The first method was to check Wi-Fi on my laptop to see what other devices, if any, appeared of the area chosen, which was my neighbourhood. I discovered another method to find Wi-Fi networks in researching ways to track how far a network's reach extends (the third step). The second method I chose to utilize was to use the Airport Utility app, which is an app that can be "used to measure signal strength" on iOS devices (Nield, 2022). This app also shows the signal strength for "all Wi-Fi networks your device can detect" (Nield, 2022). I downloaded the app on my iPhone to perform the scans. I had to reconfigure some of the settings in the app to allow for the Wi-Fi Scanner to activate (see the screenshot below).



After this was done, I was then able to go to the Airport Utility App and in the top right corner there is a "Wi-Fi Scan button" that I selected (see image below).



I ran the Wi-Fi scan and documented the results through screenshots included in the results section below with my analysis. While analysing the results, I had difficulty determining what devices were assigned what IP addresses, MAC addresses, and other identifiers, so I downloaded two network mapping programs to assist with determining that. The first was NMAP, which I had used before in CIDM6340 Network Management and Information Security, and the second was an app called Fing, which I learned about in performing research on determining which MAC addresses were assigned to which devices. I was able to utilize the results from these scans to identify and verify which devices were assigned which IP addresses, which allowed me to understand which devices were communicating with each other. I have provided the results from both scans in the results section below.

For the second requirement, I found that there was only one open network which was titled “onn 5_1_2 SOUNDBAR...” I first ran a Wireshark scan on my personal network “vexus is better” for 30 seconds and included the results and analysis in the section below. I then ran a Wireshark scan on the Soundbar network for 39 seconds (the intention was to only run it for 30 seconds; however, the scan was not picking up packets being delivered as often as the scan done on my network). I include the results and analysis in the section below. In my research and analysis of the results, I also ran a network map scan using NMAP to identify devices on the network to better analyse the data from the Wireshark scan and confirm which devices are communicating with each other.

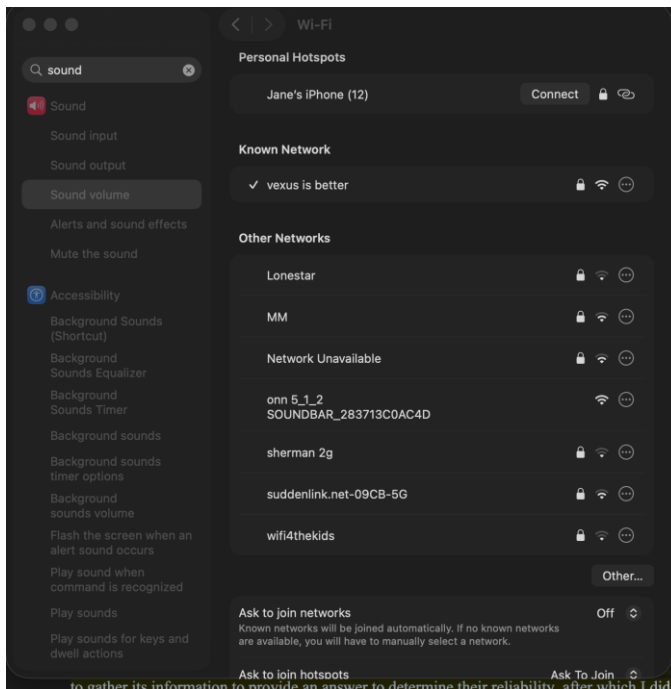
For the last requirement, I once again utilized the Airport Utility app. I checked what the dBm was initially when opened in my living room. I then walked outside through the front door and walked to the east of the house, checking the scan every so often to determine when the Wi-Fi would no longer be usable. Some research was done prior to this step to ensure that I would be able to know for sure when I hit the point that I would no longer be able to use the Wi-Fi. Once I determined the distance, I headed back to the house, then went North. I could not go west of the house, as there is a pond that I would have had to walk into to determine a distance, and I did not go south as I had already had the distance from going north and east.

Question 2: what were the results?

Summarize and explain what your findings mean. Provide any printouts of products produced, this could be a report, a map, a table, etc.

The results for the Wireshark assignment are shown below:

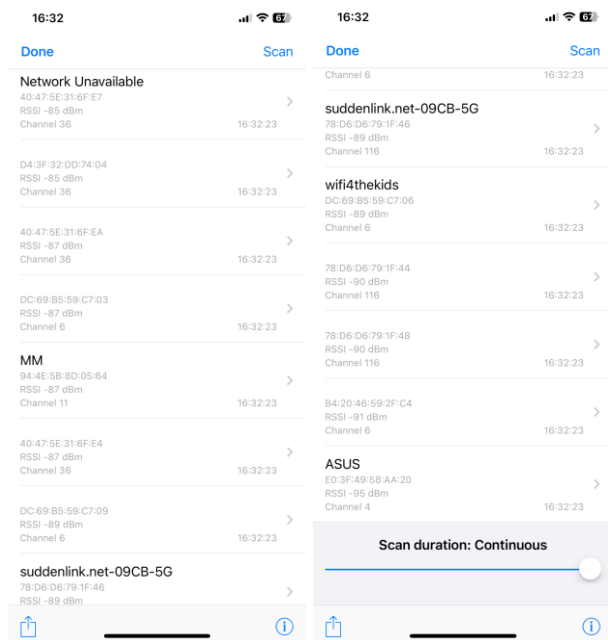
- 1) For the first part, the survey of an area (such as a neighbourhood or an apartment complex) I surveyed my neighbourhood and the surrounding networks in two ways. The first way was by looking at the Wi-Fi networks that appeared on my laptop. The image below shows the results:



As shown, it is indicated that most of the Wi-Fi networks that appear as available are password protected. There is only one network that is not password protected, and it is the “onn 5_1_2 SOUNDBAR...” network.

I also searched the nearby available networks by using the Airport Utility app. The images below show the results that occurred because of this search. Many of the same networks that were listed by my laptop through the Wi-Fi available networks were also found by my phone through the Airport Utility app and are listed below.

16:32	16:32	16:32
Done	Scan	Done
vexus is better 28-EC-22:4F:02:46 RSSI -46 dBm Channel 11	62:45:B7:37:12:A1 RSSI -57 dBm Channel 44	40:47:5E:31:6F:E3 RSSI -78 dBm Channel 6
28-EC-22:4F:02:49 RSSI -47 dBm Channel 11	vexus is better 28-EC-22:4E:9C:46 RSSI -57 dBm Channel 11	Network Unavailable 40:47:5E:31:6F:E6 RSSI -78 dBm Channel 6
28-EC-22:4F:02:43 RSSI -48 dBm Channel 11	28-EC-22:4E:9C:44 RSSI -58 dBm Channel 36	suddenlink.net-09CB-5G 78:D6:D6:79:1F:45 RSSI -81 dBm Channel 11
28-EC-22:4F:02:44 RSSI -51 dBm Channel 36	28-EC-22:4E:9C:4A RSSI -58 dBm Channel 36	78:D6:D6:79:1F:43 RSSI -82 dBm Channel 11
vexus is better 28-EC-22:4F:02:47 RSSI -51 dBm Channel 36	vexus is better 28-EC-22:4E:9C:47 RSSI -58 dBm Channel 36	78:D6:D6:79:1F:47 RSSI -82 dBm Channel 11
28-EC-22:4F:02:4A RSSI -51 dBm Channel 36	Network Unavailable D4:3F:32:DD:74:06 RSSI -75 dBm Channel 6	D4:3F:32:DD:74:0A RSSI -84 dBm Channel 36
onn 5_1_2 SOUNDBAR_283713C0AC4D 16:D4:24:47:80:07 RSSI -52 dBm Channel 2	40:47:5E:31:6F:E9 RSSI -78 dBm Channel 6	Network Unavailable D4:3F:32:DD:74:07 RSSI -84 dBm Channel 36
62:45:B7:37:12:A1 RSSI -57 dBm Channel 44	40:47:5E:31:6F:E3 RSSI -78 dBm Channel 6	Network Unavailable D4:3F:32:DD:74:07 RSSI -84 dBm Channel 36



- 2) For the second request, some of the open networks that were detected be analysed, or your own if you wanted to be safe. It was specifically requested that the protocols and other specific information that Wireshark offers be analysed. For this request, I scanned the “onn 5_1_2 SOUNDBAR...” network and our personal network “vexus is better.” For the SOUNDBAR network, I ran the scan for 39 seconds and received 31 lines of packet data. There were only two sources from which the data was being received. One was the IP address 192.168.255.21, which was determined to be the router/gateway device and the second one was at the IPv6 address fe80::1067:92b7:efd6:4ef9 which was determined to be the soundbar itself. I was able to confirm this using NMAP scans and by performing further research. My research results stated that the “fe80:” start to the IPv6 address is often reserved for “link-local” addresses, which are used for communications within a “single network segment or link” (DumpsQueen, n.d.). These addresses can only communicate with one another within a singular network, which means that this would have to be the soundbar itself. Along with the sources, there were three different destinations that were scanned. The first was the DNS address 224.0.0.25, which only received packets from the IP address 192.168.255.21. The second destination was ff02::2, the third was ff02::fb, and the fourth was ff02::16. These three destinations only ever received packets from the IPv6 address fe80::1067:92b7:efd6:4ef9 because, as previously stated, they are link local addresses communicating with each other within the network. The scan consisted of ICMPv6 (internet control message protocol version 6) and MDNS (multicast domain name system) protocols. This indicates that these destinations were working together to manage the flow of traffic between the soundbar and the device(s) attempting to connect to it, in this case was my MacBook.

For the Wireshark scan of my personal network, there was 2,555 lines of data that were tracked in the 30 second scan. There were several different protocols that were

used throughout this scan, which include ARP, DNS, ICMPv6, MDNS, QUIC, TCP, TLSv1.2, TLSv1.3, and UDP protocols. These protocols performed allow for different functions to be performed. ARP, also known as Address Resolution Protocol, is a function used to map “logical addresses to the physical addresses” in a network ([10] Geeksforgeeks, 2024). This protocol can be seen being used throughout the scan when devices are trying to identify where certain IP addresses were at. The IP address 192.168.4.1 was the most common asked about IP address. It was first seen in line 4, where it identified that the address 192.168.4.1 is located at 28:ec:22:4e:9c:52. It is then seen at various points throughout the remainder of the scan, when devices request the devices at specific IP addresses to identify themselves. The devices that were communicating most through this protocol were devices that were devices that were broadcasting something, such as the Ring Cameras (identified below by their IP addresses and the source – TexasInstrum – or the Visio TV. An image of all ARP scan results can be seen below:

No.	Time	Source	Destination	Protocol	Length	Info
4	0.308828	eero_4e:9c:52	2a:a0:02:3a:d8:49	ARP	42	192.168.4.1 is at 28:ec:22:4e:9c:52
35	1.106873	TexasInstrum_f2:8f...	Broadcast	ARP	74	ARP Announcement for 192.168.4.37
150	3.769201	TexasInstrum_f2:8f...	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.32
229	3.972892	Vizio_97:e3:d7	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.35
590	5.004224	UniversalGlo_ad:a6...	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.24
1090	11.959725	Vizio_97:e3:d7	Broadcast	ARP	74	Who has 192.168.4.35? (ARP Probe)
1091	11.959727	TexasInstrum_f2:d8...	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.33
1119	15.033662	Vizio_97:e3:d7	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.35
1532	16.983392	eero_4e:9c:52	2a:a0:02:3a:d8:49	ARP	42	Who has 192.168.4.67? Tell 192.168.4.1
1533	16.983504	2a:a0:02:3a:d8:49	eero_4e:9c:52	ARP	42	192.168.4.67 is at 2a:a0:02:3a:d8:49
2022	22.199804	TexasInstrum_2e:33...	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.34
2041	23.224217	TexasInstrum_19:76...	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.25
2065	26.092222	Vizio_97:e3:d7	Broadcast	ARP	74	Who has 192.168.4.1? Tell 192.168.4.35

DNS, also known as domain name system, is a naming system that translates domain names into IP addresses ([11] GeeksforGeeks, 2024). The DNS process is utilized so that the user does not have to remember the IP addresses for each individual website they want to visit as the DNS process does that for the user. The scan below shows that this protocol was used most by my MacBook (IP 192.168.4.67) communicating with various websites such as “wtamu” and “word-edit.officeapps” that can be read in part from the information section. The DNS protocol results from the scan are shown below:

83	3.180596	192.168.4.67	216.167.161.37	DNS	91	Standard query 0x9834 HTTPS signaler-pa.clients6.google.com
84	3.180751	192.168.4.67	216.167.161.37	DNS	91	Standard query 0x5031 A signaler-pa.clients6.google.com
85	3.205828	216.167.161.37	192.168.4.67	DNS	141	Standard query response 0x9834 HTTPS signaler-pa.clients6.google.com 50...
86	3.205830	216.167.161.37	192.168.4.67	DNS	107	Standard query response 0x5031 A signaler-pa.clients6.google.com A 142...
1060	11.630198	192.168.4.67	216.167.161.37	DNS	81	Standard query 0xf6f9 HTTPS wtamu.instructure.com
1061	11.630335	192.168.4.67	216.167.161.37	DNS	81	Standard query 0x6899 A wtamu.instructure.com
1062	11.650625	216.167.161.37	192.168.4.67	DNS	257	Standard query response 0xf6f9 HTTPS wtamu.instructure.com CNAME cluste...
1063	11.650628	216.167.161.37	192.168.4.67	DNS	223	Standard query response 0x6899 A wtamu.instructure.com CNAME cluster384...
1539	17.620939	192.168.4.67	216.167.161.37	DNS	93	Standard query 0x17c1 HTTPS usc-word-edit.officeapps.live.com
1540	17.621067	192.168.4.67	216.167.161.37	DNS	93	Standard query 0xcc78 A usc-word-edit.officeapps.live.com
1542	17.633855	216.167.161.37	192.168.4.67	DNS	291	Standard query response 0xcc78 A usc-word-edit.officeapps.live.com CNAM...
1543	17.633857	216.167.161.37	192.168.4.67	DNS	316	Standard query response 0x17c1 HTTPS usc-word-edit.officeapps.live.com ...
2531	30.316372	192.168.4.67	216.167.161.37	DNS	84	Standard query 0x4e87 HTTPS wtamu.hosted.panopto.com
2532	30.316538	192.168.4.67	216.167.161.37	DNS	84	Standard query 0x06a2 A wtamu.hosted.panopto.com
2533	30.350349	216.167.161.37	192.168.4.67	DNS	235	Standard query response 0x4e87 HTTPS wtamu.hosted.panopto.com CNAME hos...
2534	30.350352	216.167.161.37	192.168.4.67	DNS	249	Standard query response 0x06a2 A wtamu.hosted.panopto.com CNAME hosted...

ICMPv6, also known as internet control message protocol version 6, is a network protocol that is used to forward “operational information and error messages” to other devices when there are errors, diagnostic messages, or information bottlenecks in communications within a network ([10] GeeksforGeeks, 2024). That was seen in the network scan being used to send messages internally between devices to show when a destination port was unreachable (line 30) or when there were messages such as

advertisements and solicitations (line 596, 597, 607, and 608). A screenshot of the scan results that utilized the ICMPv6 protocol are shown below:

30	1.011438	fe80::c53:c03f:e8e...	fe80::184e:f2a7:b061:cd73	ICMPv6	114 Destination Unreachable (Port unreachable)
596	5.430776	fe80::184e:f2a7:b0...	fe80::c53:c03f:e8ef:ef7a	ICMPv6	86 Neighbor Solicitation for fe80::c53:c03f:e8ef:ef7a from 34:fd:6a:33:c4:...
597	5.431033	fe80::c53:c03f:e8e...	fe80::184e:f2a7:b061:cd73	ICMPv6	78 Neighbor Advertisement fe80::c53:c03f:e8ef:ef7a (sol)
607	6.065003	fe80::c53:c03f:e8e...	fe80::184e:f2a7:b061:cd73	ICMPv6	86 Neighbor Solicitation for fe80::184e:f2a7:b061:cd73 from 2a:a0:02:3a:d8...
608	6.077484	fe80::184e:f2a7:b0...	fe80::c53:c03f:e8ef:ef7a	ICMPv6	78 Neighbor Advertisement fe80::184e:f2a7:b061:cd73 (sol)

mDNS, also known as multicast domain name system, is used within a small network to “resolve hostnames to IP addresses” (Wikipedia, n.d.). It was made to be used in tandem with DNS protocols or by itself and is largely used by Apple “Bonjour” software. This can be seen in the scan that was done below:

2483	28.159428	192.168.4.67	224.0.0.251	MDNS	116 Standard query 0x0000 SRV Jane's AirPort Time Capsule._afpovertcp._tcp...
2484	28.159463	fe80::c53:c03f:e8e...	ff02::fb	MDNS	136 Standard query 0x0000 SRV Jane's AirPort Time Capsule._afpovertcp._tcp...
2492	29.162506	192.168.4.67	224.0.0.251	MDNS	116 Standard query 0x0000 SRV Jane's AirPort Time Capsule._afpovertcp._tcp...
2493	29.162527	fe80::c53:c03f:e8e...	ff02::fb	MDNS	136 Standard query 0x0000 SRV Jane's AirPort Time Capsule._afpovertcp._tcp...

QUIC, also known as quick UDP internet connections, is a transport layer protocol that is supposed to make web communications “faster and more reliable” by working over UDP protocols rather than TCP protocols ([12] GeeksforGeeks, 2025). The QUIC protocol results are shown below:

87	3.207327	192.168.4.67	142.251.116.95	QUIC	1292 Initial, DCID=6efe59816a985084, PKN: 1, CRYPTO, PING, CRYPTO, CRYPTO, C...
88	3.207356	192.168.4.67	142.251.116.95	QUIC	1292 Initial, DCID=6efe59816a985084, PKN: 2, CRYPTO
89	3.207393	192.168.4.67	142.251.116.95	QUIC	1292 Initial, DCID=6efe59816a985084, PKN: 3, PADDING, CRYPTO, CRYPTO, CRYPTO...
90	3.207729	192.168.4.67	142.251.116.95	QUIC	121 0-RTT, DCID=6efe59816a985084
91	3.209099	192.168.4.67	142.251.116.95	QUIC	588 0-RTT, DCID=6efe59816a985084
92	3.247577	142.251.116.95	192.168.4.67	QUIC	82 Initial, SCID=eefe59816a985084, PKN: 1, ACK
93	3.247580	142.251.116.95	192.168.4.67	QUIC	82 Initial, SCID=eefe59816a985084, PKN: 2, ACK
94	3.247581	142.251.116.95	192.168.4.67	QUIC	1292 Initial, SCID=eefe59816a985084, PKN: 3, ACK, PADDING
95	3.247583	142.251.116.95	192.168.4.67	QUIC	1292 Initial, SCID=eefe59816a985084, PKN: 4, ACK, PADDING
96	3.253277	142.251.116.95	192.168.4.67	QUIC	1292 Initial, SCID=eefe59816a985084, PKN: 5, CRYPTO, PADDING
97	3.253281	142.251.116.95	192.168.4.67	QUIC	343 Protected Payload (KP0)
98	3.253282	142.251.116.95	192.168.4.67	QUIC	991 Protected Payload (KP0)
99	3.253284	142.251.116.95	192.168.4.67	QUIC	92 Protected Payload (KP0)
100	3.253286	142.251.116.95	192.168.4.67	QUIC	66 Protected Payload (KP0)
101	3.253963	192.168.4.67	142.251.116.95	QUIC	120 Handshake, DCID=eefe59816a985084
102	3.254018	192.168.4.67	142.251.116.95	QUIC	73 Protected Payload (KP0), DCID=eefe59816a985084
103	3.277245	142.251.116.95	192.168.4.67	QUIC	80 Protected Payload (KP0)
104	3.277248	142.251.116.95	192.168.4.67	QUIC	63 Protected Payload (KP0)
105	3.277569	192.168.4.67	142.251.116.95	QUIC	73 Protected Payload (KP0), DCID=eefe59816a985084
106	3.289531	142.251.116.95	192.168.4.67	QUIC	162 Protected Payload (KP0)
107	3.289534	142.251.116.95	192.168.4.67	QUIC	225 Protected Payload (KP0)
108	3.289858	192.168.4.67	142.251.116.95	QUIC	73 Protected Payload (KP0), DCID=eefe59816a985084
110	3.316296	192.168.4.67	142.251.116.95	QUIC	78 Protected Payload (KP0), DCID=eefe59816a985084
113	3.353000	142.251.116.95	192.168.4.67	QUIC	65 Protected Payload (KP0)

The TCP, or transmission control protocol, is used to transmit data over a network and establish a connection between devices before sending data ([10] GeekforGeeks, 2024). This was the most common protocol that was identified in the Wireshark scan, and was seen most between my MacBook an outside server. Some of the results are shown below:

2428	27.858752	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=401413 Ack=2748 Win=72704 Len=1428 TSval=145...
2429	27.858753	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=402841 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2430	27.858754	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=404269 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2431	27.858755	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=405697 Ack=2748 Win=72704 Len=1428 TSval=145...
2432	27.858756	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=407125 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2433	27.858757	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=408553 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2434	27.858758	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=409981 Ack=2748 Win=72704 Len=1428 TSval=145...
2435	27.858844	192.168.4.67	3.170.115.107	TCP	66	53780 → 443	[ACK] Seq=2748 Ack=411409 Win=496640 Len=0 TSval=123639632 ...
2436	27.871440	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=411409 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2438	27.871457	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=414265 Ack=2748 Win=72704 Len=1428 TSval=145...
2439	27.871459	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=415693 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2440	27.871460	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=417121 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2441	27.871462	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=418549 Ack=2748 Win=72704 Len=1428 TSval=145...
2442	27.871463	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=419977 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2443	27.871465	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=421405 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2444	27.871467	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=422833 Ack=2748 Win=72704 Len=1428 TSval=145...
2445	27.871468	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=424261 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2446	27.871469	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=425689 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2447	27.871470	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=427117 Ack=2748 Win=72704 Len=1428 TSval=145...
2448	27.871472	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=428545 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2450	27.871475	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=431401 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2451	27.871476	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=432829 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2452	27.871477	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=434257 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2453	27.871478	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=435685 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2454	27.871480	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[PSH, ACK] Seq=437113 Ack=2748 Win=72704 Len=1428 TSval=145...
2455	27.871481	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=438541 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2456	27.871482	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=439969 Ack=2748 Win=72704 Len=1428 TSval=14550215...
2457	27.871484	3.170.115.107	192.168.4.67	TCP	1494	443 → 53780	[ACK] Seq=441397 Ack=2748 Win=72704 Len=1428 TSval=14550215...

TLSv1, also known as transport layer security, is used to send secure, encrypted packets between “servers and web apps” ([10] GeeksforGeeks, 2024). The TLSv1 packet transactions can be seen below:

9	0.385047	192.168.4.67	44.206.27.227	TLSv1..	569	Client Hello (SNI=wtamu.hosted.panopto.com)
11	0.448200	44.206.27.227	192.168.4.67	TLSv1..	300	Server Hello, Change Cipher Spec, Application Data, Application Data
13	0.449058	192.168.4.67	44.206.27.227	TLSv1..	130	Change Cipher Spec, Application Data
14	0.449323	192.168.4.67	44.206.27.227	TLSv1..	158	Application Data
17	0.449721	192.168.4.67	44.206.27.227	TLSv1..	327	Application Data
18	0.449780	192.168.4.67	44.206.27.227	TLSv1..	467	Application Data
21	0.511641	44.206.27.227	192.168.4.67	TLSv1..	245	Application Data
22	0.511643	44.206.27.227	192.168.4.67	TLSv1..	137	Application Data
24	0.512845	192.168.4.67	44.206.27.227	TLSv1..	97	Application Data
25	0.539862	44.206.27.227	192.168.4.67	TLSv1..	1003	Application Data
26	0.539864	44.206.27.227	192.168.4.67	TLSv1..	97	Application Data
44	1.784686	192.168.4.67	52.108.36.35	TLSv1..	517	Client Hello (SNI=word-telemetry.officeapps.live.com)
46	1.938933	52.108.36.35	192.168.4.67	TLSv1..	153	Hello Retry Request, Change Cipher Spec
48	1.940883	192.168.4.67	52.108.36.35	TLSv1..	803	Change Cipher Spec, Client Hello (SNI=word-telemetry.officeapps.live.co...
53	2.104189	52.108.36.35	192.168.4.67	TLSv1..	1514	Server Hello
57	2.104199	52.108.36.35	192.168.4.67	TLSv1..	1168	Application Data
60	2.129141	192.168.4.67	52.108.36.35	TLSv1..	128	Application Data
61	2.129293	192.168.4.67	52.108.36.35	TLSv1..	146	Application Data
63	2.129507	192.168.4.67	52.108.36.35	TLSv1..	718	Application Data
64	2.129529	192.168.4.67	52.108.36.35	TLSv1..	915	Application Data
68	2.291496	52.108.36.35	192.168.4.67	TLSv1..	157	Application Data
69	2.291497	52.108.36.35	192.168.4.67	TLSv1..	116	Application Data
71	2.292091	192.168.4.67	52.108.36.35	TLSv1..	85	Application Data
74	2.337724	44.206.27.227	192.168.4.67	TLSv1..	90	Application Data
77	2.446637	52.108.36.35	192.168.4.67	TLSv1..	1012	Application Data, Application Data
120	3.685212	192.168.4.67	52.108.36.35	TLSv1..	79	Application Data
122	3.685237	192.168.4.67	52.108.36.35	TLSv1..	425	Application Data
128	3.714395	192.168.4.67	52.108.8.12	TLSv1..	895	Client Hello (SNI=common.online.office.com)
134	3.730568	52.108.8.12	192.168.4.67	TLSv1..	153	Hello Retry Request, Change Cipher Spec
136	3.731306	192.168.4.67	52.108.8.12	TLSv1..	1181	Change Cipher Spec, Client Hello (SNI=common.online.office.com)
139	3.751352	52.108.8.12	192.168.4.67	TLSv1..	345	Server Hello, Application Data

The UDP, also known as the user datagram protocol, is used mainly to create faster connections for multicasting and broadcasting purposes ([10] GeeksforGeeks, 2024). UDP is considered unreliable, and adds no “flow control, reliability, or error recovery” functions to packet transmissions ([10] GeeksforGeeks, 2024). Some of the UDP packets can be seen in the results below:

29	1.011300	fe80::184e:f2a7:b0...	fe80::c53:c03f:e8ef:ef7a	UDP	66	3722 → 3722	Len=4
31	1.028139	192.168.4.67	173.194.208.95	UDP	1292	57582 → 443	Len=1250
32	1.062876	173.194.208.95	192.168.4.67	UDP	1292	443 → 57582	Len=1250
33	1.071561	173.194.208.95	192.168.4.67	UDP	67	443 → 61839	Len=25
34	1.071941	192.168.4.67	173.194.208.95	UDP	74	61839 → 443	Len=32
39	1.660954	192.168.4.67	142.251.186.103	UDP	71	58853 → 443	Len=29
40	1.721786	142.251.186.103	192.168.4.67	UDP	66	443 → 58853	Len=24
79	3.168173	173.194.208.95	192.168.4.67	UDP	160	443 → 61839	Len=118
80	3.168176	173.194.208.95	192.168.4.67	UDP	64	443 → 61839	Len=22
81	3.168178	173.194.208.95	192.168.4.67	UDP	160	443 → 61839	Len=118
82	3.169025	192.168.4.67	173.194.208.95	UDP	76	61839 → 443	Len=34
109	3.292371	192.168.4.67	173.194.208.95	UDP	537	61839 → 443	Len=495
111	3.336444	173.194.208.95	192.168.4.67	UDP	71	443 → 61839	Len=29
112	3.347689	192.168.4.67	173.194.208.95	UDP	75	61839 → 443	Len=33
114	3.357951	173.194.208.95	192.168.4.67	UDP	127	443 → 61839	Len=85
115	3.369077	192.168.4.67	173.194.208.95	UDP	79	61839 → 443	Len=37
116	3.411845	173.194.208.95	192.168.4.67	UDP	67	443 → 61839	Len=25
118	3.685023	192.168.4.67	173.194.208.95	UDP	71	61839 → 443	Len=29
124	3.689471	192.168.4.67	40.99.168.242	UDP	384	54281 → 443	Len=342
129	3.720294	40.99.168.242	192.168.4.67	UDP	74	443 → 54281	Len=32
130	3.720295	40.99.168.242	192.168.4.67	UDP	69	443 → 54281	Len=27
131	3.721299	192.168.4.67	40.99.168.242	UDP	79	54281 → 443	Len=37
132	3.730565	40.99.168.242	192.168.4.67	UDP	765	443 → 54281	Len=723
137	3.736876	173.194.208.95	192.168.4.67	UDP	67	443 → 61839	Len=25
149	3.761293	192.168.4.67	40.99.168.242	UDP	84	54281 → 443	Len=42
155	3.786551	40.99.168.242	192.168.4.67	UDP	71	443 → 54281	Len=29
160	3.816733	192.168.4.67	40.99.168.242	UDP	80	54281 → 443	Len=38
192	3.941214	192.168.4.67	173.194.208.95	UDP	71	61839 → 443	Len=29
297	4.008649	173.194.208.95	192.168.4.67	UDP	67	443 → 61839	Len=25
420	4.063419	192.168.4.67	173.194.208.95	UDP	1292	57582 → 443	Len=1250

I performed a second scan of network devices using NMAP. I also downloaded the app “Fing” as another type of network scanning platform to assist with identifying which devices are assigned which IP address.

The result of the NMAP scan is below:

? (192.168.4.1) at 28:ec:22:4e:9c:52 on en0 ifscope [ethernet]
 ? (192.168.4.24) at 3c:e1:a1:ad:a6:4c on en0 ifscope [ethernet]
 ? (192.168.4.25) at a8:10:87:19:76:2b on en0 ifscope [ethernet]
 ? (192.168.4.26) at 34:fd:6a:33:c4:e0 on en0 ifscope [ethernet]
 ? (192.168.4.30) at b0:ce:18:26:ad:ee on en0 ifscope [ethernet]
 ? (192.168.4.32) at 78:4:73:f2:8f:79 on en0 ifscope [ethernet]
 ? (192.168.4.33) at 78:4:73:f2:d8:db on en0 ifscope [ethernet]
 ? (192.168.4.34) at f8:36:9b:2e:33:13 on en0 ifscope [ethernet]
 ? (192.168.4.37) at 90:9a:77:fe:1a:7f on en0 ifscope [ethernet]
 ? (192.168.4.41) at 6:24:77:33:7e:33 on en0 ifscope [ethernet]
 ? (192.168.4.56) at 12:f8:c:e8:23:1c on en0 ifscope [ethernet]
 ? (192.168.4.67) at 2a:a0:2:3a:d8:49 on en0 ifscope permanent [ethernet]
 ? (192.168.4.255) at (incomplete) on en0 ifscope [ethernet]
 mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]

The result of the Fing scan are shown below:

Fing

<

OverviewToolsAccount

...

Net 28:EC:22:4E:9C:52

DevicesNetworkSecurityInternet

Automate scanning and block devices

19 devices1 hour ago

eero

192.168.4.1

eero

eero Pro 6E

IP Camera

192.168.4.23

Blink

B0:09:DA:6B:74:83

Doorbell

192.168.4.24

Ring

Video Doorbell

IP Camera

192.168.4.25

Ring

Spotlight Cam

Media Player

192.168.4.26

Apple

Apple TV (4th Generation)

Generic

192.168.4.30

Zhejiang shenghui lighting

B0:CE:18:26:AD:EE

Computer

192.168.4.31

Asus

Windows

IP Camera

192.168.4.32

Ring

Spotlight Cam

IP Camera

192.168.4.33

Ring

Spotlight Cam

IP Camera

192.168.4.34

Ring

Spotlight Cam

Television

192.168.4.35

Vizio

Chrome OS

Doorbell

192.168.4.37

Ring

Video Doorbell

Generic

192.168.4.39

Amazon

CC:9E:A2:41:11:C6

Generic

192.168.4.41

-

06:24:77:33:7E:33

Game Console

192.168.4.44

Nintendo

BC:CE:25:7D:04:9A

Wi-Fi

192.168.4.50

eero

eero

Generic

192.168.4.54

-

62:6C:BC:64:C1:56

Wi-Fi

192.168.4.50

eero

eero

Generic

192.168.4.54

-

62:6C:BC:64:C1:56

Mobile

192.168.4.56

-

12:0F:8C:E8:23:1C

iPad

192.168.4.67

Apple

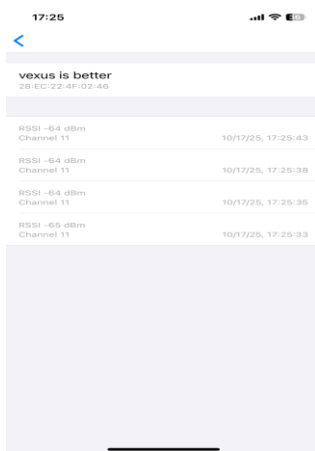
iPad Pro 12.9" 3rd Gen (Wi-Fi)

I was able to use these two different scans to identify which IP addresses were assigned to which specific devices on the network. I was also able to verify some of the IP addresses through checking the associated devices or their apps to verify they are correctly identified. I had to do this, as the Fing app data was not always accurate as to what it thought it was reading (for example, it thought that the device at 192.168.4.67 was an iPad but it really was my MacBook). The list below was generated:

IP Address:	MAC Address:	Device Name:
192.168.4.1	28:EC:22:4E:9C:52	Eero – eero pro 6E (router)
192.168.4.23	80:09:DA:6B:74:63	(Blink) Ring Camera - extender
192.168.4.24	3C:E1:A1:AD:A6:4C	Ring Camera – Front Door (Video Doorbell)
192.168.4.25	A8:10:87:19:76:2B	Ring Camera – Garage (Spotlight)
192.168.4.26	34:FD:6A:33:C4:E0	Apple TV (Bedroom TV)
192.168.4.30	B0:CE:18:26:AD:EE	Zhejiang Shenghui Lighting – smart lights
192.168.4.31	E8:9C:25:7A:0D:0F	Computer – Asus (Austin gaming computer)
192.168.4.32	78:04:73:F2:8F:79	Ring Camera – Front
192.168.4.33	78:04:73:F2:D8:DB	Ring Camera – Side
192.168.4.34	F8:36:9B:2E:33:13	Ring Camera - Side
192.168.4.35	2C:64:1F:97:E3:D7	Vizio TV (Living Room TV)
192.168.4.37	90:9A:77:FE:1A:7F	Ring Camera – Garden
192.168.4.39	CC:9E:A2:41:11:C6	Amazon – Alexa
192.168.4.41	06:24:77:33:7E:33	Mobile Phone – iPhone (Austin)
192.168.4.44	BC:CE:25:7D:04:9A	Nintendo – Game Console
192.168.4.50	28:EC:22:4F:02:52	Eero – Wi-Fi extender
192.168.4.54	62:6C:BC:64:C1:56	Playstation – Game Console
192.168.4.56	12:0F:8C:E8:23:1C	Mobile Phone – iPhone (Jane)
192.168.4.67	2A:A0:02:3A:D8:49	MacBook Pro – Computer (Jane)

- 1) For the last request, which was to check how far your own network extends to see if it can be found and accessed outside of your home, I was able to do this by using the Airport Utility app on my iPhone device. To be able to utilize the Airport Utility app appropriately, I had to perform some research on the dBm unit of measurement. The dBm unit of measurement stands for “decibels relative to 1 milliwatt,” and is a “logarithmic unit” that is used to measure “electrical or radio” systems (see reference [4]). I learned that the closer the dBm value is to 0, the stronger that the signal is, and the point at which a network becomes unusable is when the device registers the signal as being in the –80 to –90 dBm range. The ideal range would be for a device to register at –40 dBm to –60 dBm range, and which a –60 dBm to –80 dBm range shows a slower signal that will still facilitate a two-way data transfer capability. The Airport Utility app used this unit of measurement to determine the strength of the

signal that my device was receiving from the network at the current location of that device. I was able to test the physical boundaries of my network's range, as when the signal reached the point of -80dBm on the Airport Utility app, I knew that my device no longer had a strong enough connection to the Wi-Fi network to still connect to that network. The images below show the results of the scan. The first image shows the results for when I walked east of the house, as well as a photo of how far away from the house I was able to walk before I reached the results of -80 dBm , with the red arrow pointing at my house. When the test scanned at -80 dBm , I deemed this to be the range that the network is extended to. Below is the scan performed from right outside the house. The results show -64 dBm , which means that the Wi-Fi is scanned at operating at a good rate. I started here and walked East and North from my house to get an idea of how far the network can be reached from outside the house.



RSSI -64 dBm Channel 11	10/17/25, 17:25:43
RSSI -64 dBm Channel 11	10/17/25, 17:25:38
RSSI -64 dBm Channel 11	10/17/25, 17:25:35
RSSI -65 dBm Channel 11	10/17/25, 17:25:33

While I was outside performing this research on our personal network's extension, my husband came home and saw me walking around outside performing these scans. He thought I looked crazy, but when I told him what I was doing and why, he provided some information that helped explain the reason for our personal Wi-Fi network's extent. He stated that we had a Wi-Fi extender, which was helping extend our network's reach beyond what it would have been without it. He stated that he had this installed so that our Wi-Fi would reach all around the outside of the house to ensure that all our Ring security cameras were connected to the network. The images below show the scan at the point that the internet rate was tracked at -80 dBm , as well as an image of where I was in relation to the house. The red arrow is pointing to our house (which is the one on the corner).

17:23

<

vexus is better
28:EC-22:4F:02:46

RSSI -80 dBm Channel 11	10/17/25, 17:23:42
RSSI -81 dBm Channel 11	10/17/25, 17:23:31
RSSI -80 dBm Channel 11	10/17/25, 17:23:25



The next two images show the extension to the north. I followed the same process, which was to see how far away from the house I could get heading north before the scan resulted in -80 dBm. The red arrow shows which house is mine. I did not walk to the west of the house, as it would have required that I walk into the grassy marsh next to the pond that is there. I did not walk south of the house, as I felt that I had a good idea of what the distance is that the network extended to from the other two scans.

17:28

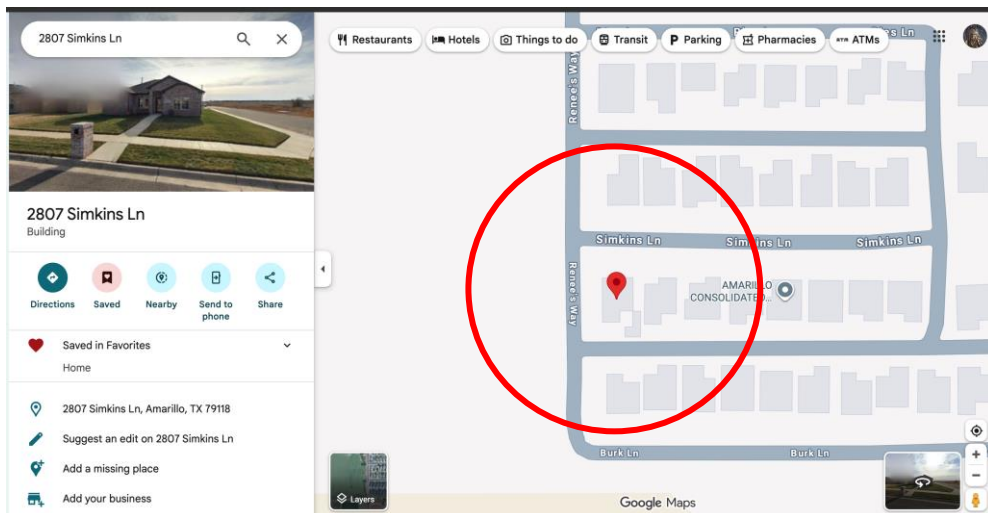
<

vexus is better
28:EC-22:4F:02:46

RSSI -82 dBm Channel 11	10/17/25, 17:28:30
RSSI -85 dBm Channel 11	10/17/25, 17:28:17
RSSI -85 dBm Channel 11	10/17/25, 17:28:11
RSSI -81 dBm Channel 11	10/17/25, 17:28:01
RSSI -80 dBm Channel 11	10/17/25, 17:27:55
RSSI -79 dBm Channel 11	10/17/25, 17:27:48
RSSI -78 dBm Channel 11	10/17/25, 17:27:43
RSSI -78 dBm Channel 11	10/17/25, 17:27:35
RSSI -75 dBm Channel 11	10/17/25, 17:27:31
RSSI -69 dBm Channel 11	10/17/25, 17:27:27
RSSI -75 dBm	



Below is a rough estimate of the Wi-Fi network's extension using Google maps to centralize the house's location. Extending the network beyond the house/property that we reside on is not the best in terms of security, as creates a greater exposure to possible security threats. This is because bad players can see the network exists and can possibly access it from further away from our house (QSFPTEK., n.d.). Another downside to having the Wi-Fi extender is that because it is extending the distance the signal is reaching, which can lead to reduced speeds and more congestion in network communications (QSFPTEK, n.d.).



Results for the SonicWall Phishing Test:

SONICWALL®
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Solutions
Partners
Support
Compi

GOOD JOB, JANE STASHI!

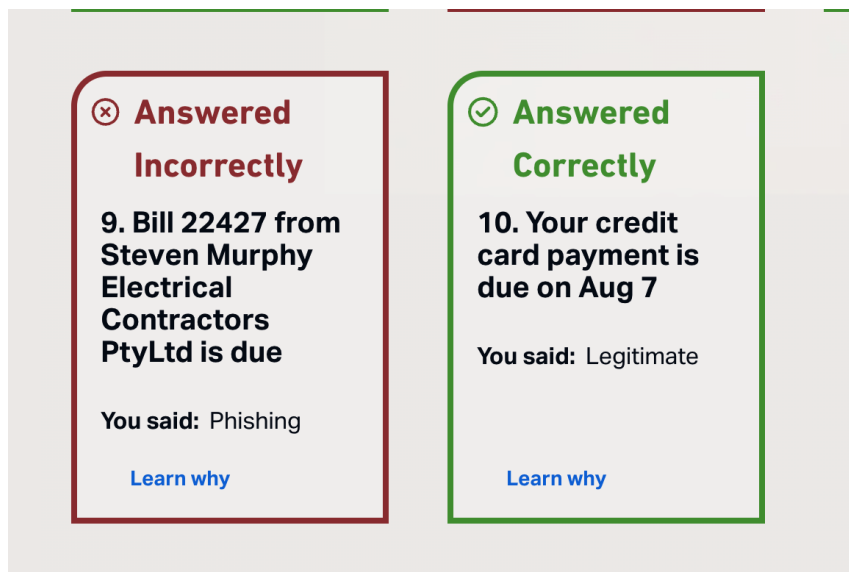
You answered **7/10** questions correctly

Email Me My Results

CHECK YOUR INBOX FOR YOUR EMAIL WITH YOUR SPECIFIC SONICWALL PHISHING QUIZ RESULTS.

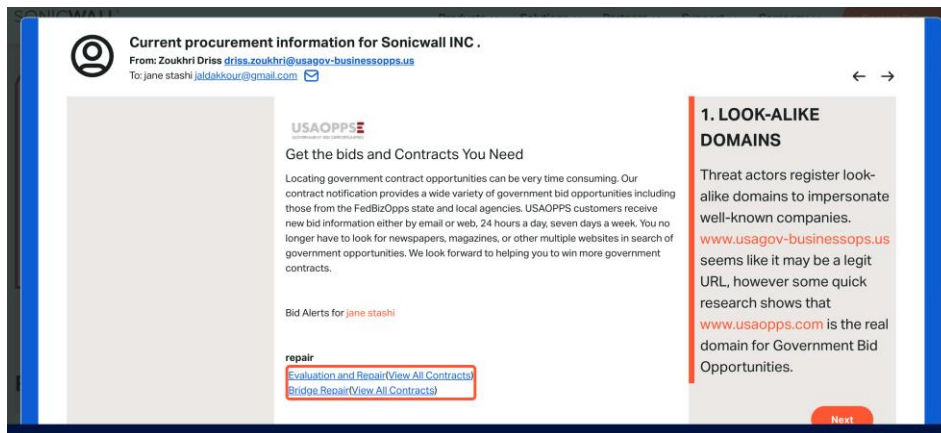
SONICWALL®
Products
Solutions
Partners
Support
Company
Contact Sales

✓ Answered Correctly 1. You've received a pricing inquiry! You said: Phishing Learn why	✓ Answered Correctly 2. Failed DHL Delivery You said: Phishing Learn why	✓ Answered Correctly 3. It's time to update your Office 365 password You said: Phishing Learn why	✓ Answered Correctly 4. Confirm your account with Doodle You said: Legitimate Learn why
✓ Answered Correctly 5. You're invited to join a webcast! You said: Legitimate	✗ Answered Incorrectly 6. Current procurement information for Sonicwall INC. You said: Legitimate	✓ Answered Correctly 7. 2: qa- Partnership 7:4 You said: Phishing	✗ Answered Incorrectly 8. Microsoft Subscription Expiry You said: Legitimate

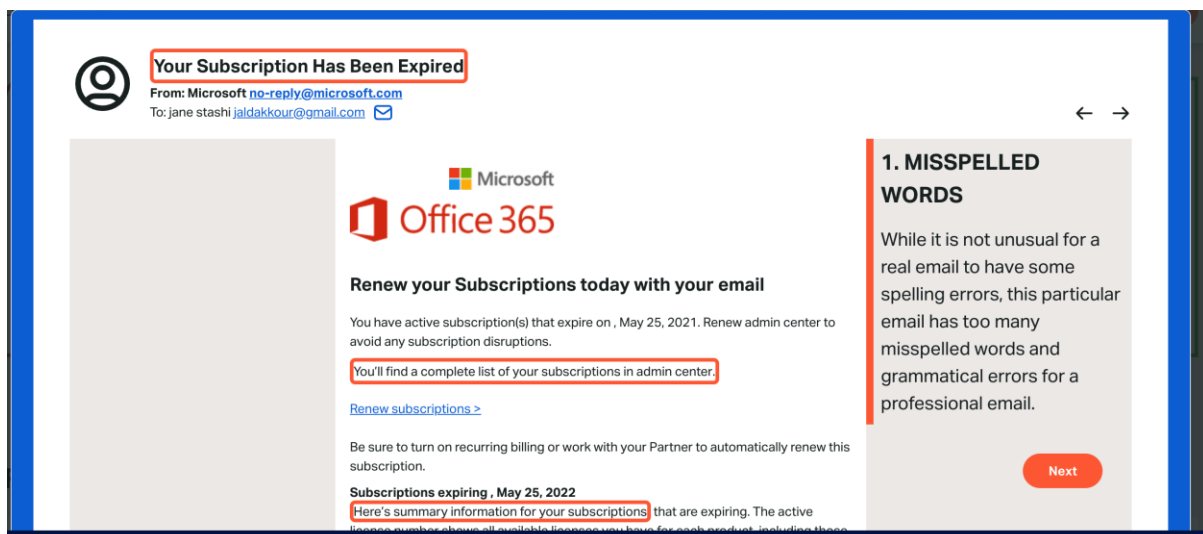


(OpenDNS, n.d.)

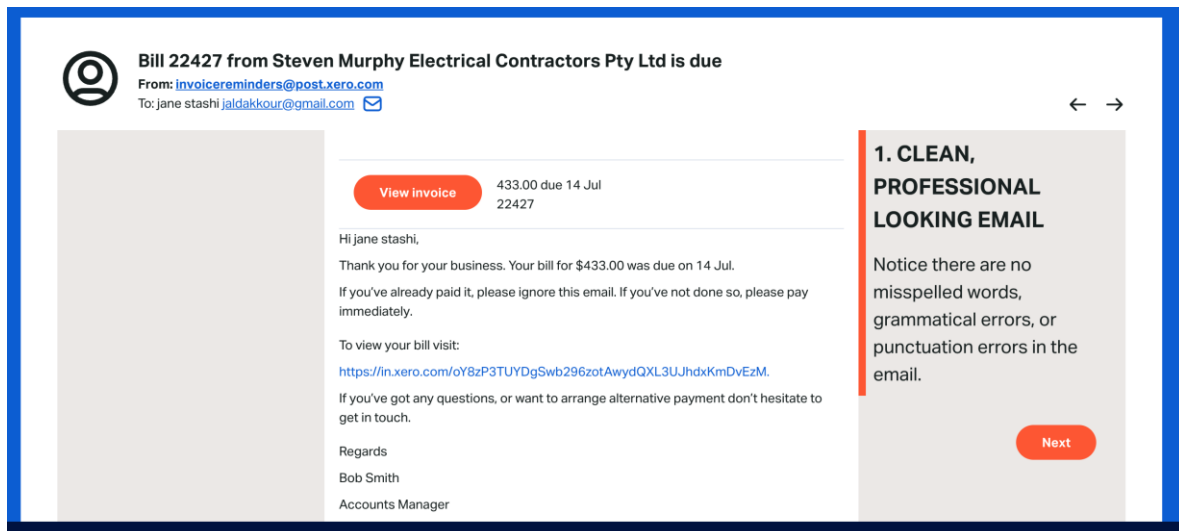
The results above are from one of the two phishing quizzes provided. My results were a 7/10. There were two phishing attempts that I fell for and said they were legitimate, and one legitimate message that I thought it was a phishing attempt. For question 6, I believed it was a legitimate email because the email account was “driss.zoukhri@usagov-businessopps.us,” which I thought was legitimate because of the “.us.” I personally have always recognized “.us” and “.gov” to be legitimate email endings typically used for or associated with US government accounts. The website offering the phishing quiz, however, stated that this is a “look-a-like domain” which is used to fool the user into thinking it is legitimate by using aspects that are assumed to be safe by users such as having a “.us” account (OpenDNS, n.d.). New research is showing that domain names ending in “.us” are “among the most prevalent” in phishing scams (Krebs, 2023). The “.us” domain is overseen by the US government and is the “country code Top Level Domain” or “ccTLD” which is a domain that each country has (e.g. Germany’s ccTLD is “.de”, which I know because when I lived in Germany, many websites such as Pinterest and Amazon had a “.de” ending to them when used in Germany) (Krebs, 2023). This is important to recognize as it makes phishing attempts that have a country’s ccTLD seem more legitimate and create a false sense of security in the user that this is a trustworthy site. Below is an image of the results and the reasons it is a phishing attempt:



The next question I got wrong was question 8, which I believed was a legitimate email, but it was a phishing scam. I believed this to be legitimate because the email was structured to look like a legitimate email from Microsoft, as well as used the Microsoft logo. This, however, was not legitimate and the tells were hidden within the message itself, as there were several misspellings and hidden hyperlinks that went to third party sites (OpenDNS, n.d.). I did not notice these as they were hidden within the body message of the email and were easily missed if not paying attention to the details of the email. Below are the images of the results and the reasons it is a phishing attempt:



The last question I got wrong was question 9, which was a legitimate email that I thought was a phishing attempt. I believed that this was a phishing attempt because I noticed that my name was in lower case "jane stashi," and so I thought it was a phishing attempt because of such. This is likely caused by how I entered my name when I started the quiz. This coupled with the lack of a company logo made me think this was a phishing attempt. Below are images of the results and the reasons it is a phishing attempt:



The results of the second phishing quiz are shown below:

PHISHING QUIZ

Think you can Outsmart Internet Scammers?

Congrats! You correctly identified 11 out of 14 sites in the OpenDNS phishing quiz. Wow, you're pretty good at spotting phish. But beware: cyber criminals are more clever than ever at creating sites that fool even the most experienced phishing detectives. Set up OpenDNS, the world's fastest-growing Internet security and DNS service, and let us take the guesswork out of identifying phishing sites. You can use OpenDNS at [home](#) or at [work](#) and be confident you're always protected, because OpenDNS automatically blocks phishing sites.

Share your results or challenge your friends:

Yahoo! — Phish **HSBC — Not a Phish** **Facebook — Not a Phish**

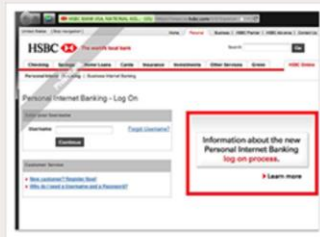
  

Yahoo! — Phish



Find out why

HSBC — Not a Phish



Facebook — Not a Phish



Twitter — Phish ❌



Find out why

American Airlines — Phish ❌



Find out why

Amazon — Not a Phish



Paypal — Phish



Find out why

Xfinity — Not a Phish



Amazon — Phish



Find out why

Hulu — Not a Phish



Ebay — Phish ❌

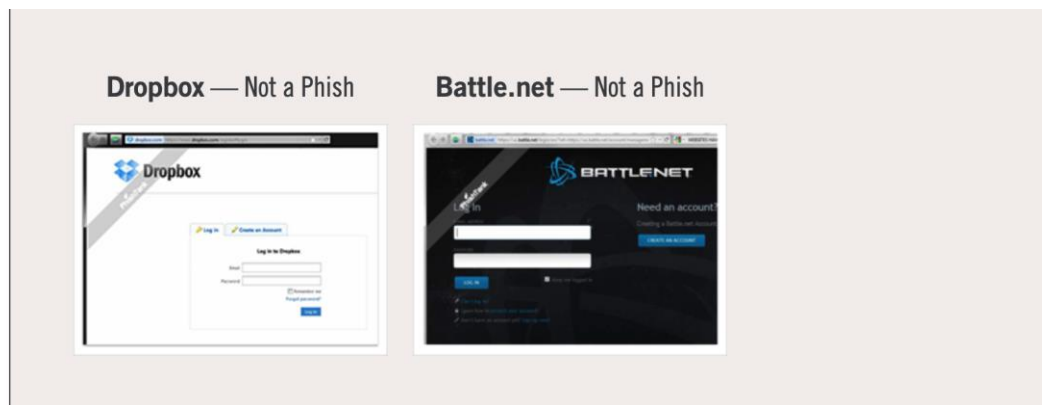


Find out why

Zynga — Phish



Find out why



I was able to correctly detect 11 out of the 14 items whether they were phishing attempts or not. The three items I was wrong about were all phishing attempts I thought were legitimate where the websites that were used as examples mimicked the websites very well. These phishing attempts could have been detected through a more thorough analysis of the URL, which would show that they are not legitimate through other means. One example of such would be that the URL starts “http” and not “https,” which signals it is secure. This, however, is not a guarantee that a website is a secure site, as it has become easier to gain a SSL certificate to “secure” a website through purchasing it from a third party (Cloudflare, n.d.). This means that the bad actors can secure an SSL certificate for their website to make it appear as a legitimate secure website when it is not.

Question 3: what did you learn?

I can't tell you what you learned. What I will say is that I reward insight. Insights are aha moments (a term in use long before Oprah wanted to copyright it). If you see new ways of doing things, new insights to your thought processes, potential future applications be they personal or work related, crossovers to other topics, these are what I reward more than just telling me you learned lots. I expect you to learn lots, but it isn't till you explain where and what that I see that you really did. Ok, so sometimes you don't learn much. I'll still grade this area high if you tell me why, what you know, how this works on what you've done in the past, etc. Sometimes when you start doing this you see that what you've learned is reinforcing what you've done and sometimes you even have small aha moments. Bottom line is to be reflective, think a few minutes or overnight about what you've done and how it fits into your nomological net (your personal set of knowledge base structure, those theories and beliefs that guide how you evaluate and use knowledge). Then write the section, when I see this done, I always score the section higher.

There were many lessons learned performing this project, even though I have used Wireshark before. One of the things I learned more about is the dBm unit of measurement. The dBm unit of measurement stands for “decibels relative to 1 milliwatt,” and is a “logarithmic unit” that is used to measure “electrical or radio” systems (see reference [4]). I learned that the closer the dBm value is to 0, the stronger that the signal is, and the point at which a network becomes unusable is when the device registers the signal as being in the -80 to -90 dBm

range. The ideal range would be for a device to register at -40 dBm to -60 dBm range, and which a -60 dBm to -80 dBm range shows a slower signal that will still facilitate a two-way data transfer capability. The app that I used – AirPort Utility – used this unit of measurement to determine the strength of the signal that my device was receiving from the network at the current location of that device. I was able to use this new knowledge to test the physical boundaries of my network's range, to learn how far it extends exactly. When the signal reached the point of -80dBm, I knew that my device no longer had a strong enough connection to the Wi-Fi network to still connect to that network. Another thing I recognized during this exercise was that the Wi-Fi bars were misleading about their connection to the network. When I hit the point where the Airport Utility app was reading in the -80 dBm range, which is when the network signal is too weak for information flow, the Wi-Fi bars on my device were still showing two full, when in my opinion they should have shown none or one to signify a significantly weaker or no connection.

Another thing this lesson drove me to learn more about was the different protocols used in device communications. From previous courses, I have learned more about network protocols such as ARP, DNS, ICMPv6, mDNS, QUIC, TCP, TLSv1, and UDP and how each one is different but necessary for certain applications or devices to communicate. I learned that certain protocols such as mDNS and ARP are utilized solely for use internally within a network, whereas others are used for communication to external servers, websites, and devices.

This lesson also taught me more about my own personal network. Performing the scans to see how far my personal network extends taught me about devices I was not aware of within our network, such as our Wi-Fi extender. Austin was the one to set up much of our network in terms of working with the network provider to set up the devices as well as maintaining the network. I knew we had a device that worked to extend a network of some kind, but I thought that it was solely related to the Ring cameras, and not to our entire network's Wi-Fi. Another thing I learned about our personal network is that our soundbar (the Onn soundbar) generates its own internal wireless network that it connects to, to connect to the “wireless subwoofer and satellite speakers” (see reference 6). I learned that it's reason for doing this is to reduce lag between the soundbar and its speakers so that the audio and visual stays in sync with each other (see reference 6). I was not aware that it was doing this, nor was Austin. Upon learning this, I've realized this could be a potential security threat for several reasons, as the soundbar can be used to find a way into our network, as it communicates with devices on our network such as our Visio TV.

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